

FERNANDO JR. H. BANAL

COMPUTER ENGINEER | SOFTWARE & SYSTEMS DEVELOPMENT

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PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on experience in full-stack development, embedded systems, networking, and technical support. Developed HR systems, document management platforms, and IoT-based research projects. Recognized through published research and award-winning technical innovations. Seeking an entry-level software, systems, or IT role where I can contribute technical and problem-solving skills.

TECHNICAL SKILLS

- Programming : Python, JavaScript, TypeScript, C++, HTML, CSS, SQL
- Frameworks & Tools: React, Bootstrap, ASP.NET Core, Git, VS Code, SSMS
- Database: SQL Server, PostgreSQL, Local Storage APIs. MongoDB
- Networking & Systems: Switch Configuration, FDAS, CCTV Systems, Tailscale, troubleshooting, technical support
- Embedded Systems: Arduino, RFID Systems, Sensors, IoT Integration. ESP Systems
- Workflow Experience: Document management, file organization, approval flows, version control support, UI/UX documentation

EXPERIENCE

Freelance Full-Stack & Embedded Systems Developer | Self-Employed | February 2025 – Present

- Developed websites and web applications using HTML, CSS, and JavaScript.
- Built embedded systems and IoT solutions using Arduino, ESP32, and C/C++.
- Integrated microcontrollers with desktop, mobile, and web-based software for real-time monitoring and control.
- Designed, tested, and optimized software and hardware solutions based on client requirements.
- Utilized Git, GitHub, APIs, databases, and cloud services to support development and deployment.

Technologies: HTML, CSS, JavaScript, Node.js, Arduino, ESP32, C/C++, Git, AWS, APIs, Databases

Systems Development and Technical Support Intern | JREMD Technologies, Inc. | Mar 2026 - Apr 2026

- Developed responsive frontend enhancements for a Document and Quotation Management System using HTML, CSS, and JavaScript.
- Implemented file management and quotation-tracking features to improve internal workflow efficiency.
- Configured remote access and deployment environments using Tailscale.
- Managed switch configuration, internet troubleshooting, and technical equipment setup for office operations.
- Collaborated with developers and technical staff in both software and on-site support tasks.

Computer Engineer Intern | Centro Escolar University | March 2026

- Optimized laboratory operations and technical administrative support for BSIT-related activities.
- Prepared and updated laboratory manuals, documentation, and instructional materials.
- Organized digital learning resources and technical files for faculty and student use.
- Provided basic troubleshooting and technical support for laboratory systems and equipment.
- Supported academic technology initiatives and documentation workflows.

Volleyball Coach & Program Developer | Passion Sports Philippines | Apr 2023 – Present

- Designed structured volleyball training programs focused on athlete development and performance improvement.
- Demonstrates leadership, communication, and the ability to coach players toward consistent process adoption.

Legal Administrative Coordinator | Iguidez-Onida Law | Mar 2019 – Sep 2019

- Managed confidential legal documents and client information with close attention to organization and security.
- Coordinated schedules, client communications, and case-file organization using digital management practices.

EDUCATION

- Bachelor of Science in Computer Engineering — Centro Escolar University | Aug 2022 – 2026
- Coursework in BS Civil Engineering — Pamantasan ng Lungsod ng Maynila | Jun 2018 – 2020

AWARDS, PUBLICATIONS, & CERTIFICATIONS

- 1st Place — Oral/Paper Competition, Search for Best Undergraduate Research 2025 | 1st Runner-Up — 17th Annual University Research Forum
- Published research in the International Journal of Science Architecture Technology and Environment (DOI: 10.63680/ijsate0226017.017).
- Published researched on LMS page views and board-exam outcomes in the Global Journal of Arts, Humanities and Social Sciences
- Selected Talent Nationwide, UNLEASH Philippines Innovation Lab Talent Program — Royal Academy of Engineering & UNLEASH Philippines (2025)
- Technical credentials include Fortinet cybersecurity certifications, EnGenius wireless/network certifications, and coursework in embedded machine learning, IoT boards, programming, and FPGA systems

SELECTED PROJECTS

Document & Quotation Management System | Apr 2026

- Developed a web-based document and quotation management system to streamline internal business operations.
- Implemented document organization, quotation tracking, and centralized record management features.
- Improved document accessibility and workflow efficiency through an intuitive user interface.
- Utilized modern web technologies to support secure and efficient document processing.

RFID-Enabled Autonomous Inventory Management System | Research Project | March 2025

- Designed and developed a 4-wheeled autonomous robot equipped with a UHF RFID scanner for real-time inventory monitoring.
- Built a web-based inventory management application that receives and displays RFID scan data in real time.
- Integrated embedded hardware, RFID technology, and software systems to automate inventory tracking and reporting.
- Implemented obstacle avoidance and autonomous navigation capabilities for efficient warehouse and inventory operations.
- Deployed the inventory management platform on AWS EC2 to enable centralized access and remote monitoring.
- Reduced manual inventory checking efforts through automation and real-time asset visibility.

SpeedTyping Learning Platform | March 2026

- Developed a web-based typing improvement platform featuring structured lessons and typing exercises.
- Implemented real-time Words Per Minute (WPM) and accuracy tracking to measure user performance.
- Designed progressive learning modules that help users improve typing speed, accuracy, and keyboard familiarity.
- Enhanced user productivity by providing guided practice sessions and performance analytics.